United States Patent Application Publication	20250279345
Kind Code	A1
Publication Date	September 04, 2025
Inventor(s)	RAGHURAMAN; Praveen

APPARATUS AND METHODS FOR DIE INTERCONNECT ARCHITECTURES AND PACKAGING

Abstract

Methods and apparatuses are provided to improve pin utilization within die architectures. In one example, a die package includes a first pin, a second pin, a first die, a second die, and a third die. The first die is electrically coupled to the first pin over a first interconnect, and to the second pin over a second interconnect. The first die is also electrically coupled to the second die over a third interconnect, and to the third die over a fourth interconnect. The first die is configured to receive a first signal over the first interconnect. Based on the first signal, the first die is configured to electrically connect the second interconnect to either the third interconnect, or the fourth interconnect. A second signal is received over the second interconnect, and is transmitted to the one of the third interconnect and the fourth interconnect electrically connected to the second interconnect.

Inventors: **RAGHURAMAN; Praveen (Chennai, IN)**

Applicant: **QUALCOMM Incorporated** (San Diego, CA)

Family ID: **96880540**

Appl. No.: **18/592818**

Filed: **March 01, 2024**

Publication Classification

Int. Cl.: **H01L23/498** (20060101); **H01L23/00** (20060101); **H01L25/065** (20230101)

U.S. Cl.:

CPC **H01L23/49838** (20130101); **H01L23/49816** (20130101); **H01L24/08** (20130101); **H01L24/48** (20130101); **H01L25/0652** (20130101); H01L2224/08145 (20130101); H01L2224/08155 (20130101); H01L2224/48149 (20130101); H01L2224/48155

(20130101); H01L2924/1431 (20130101); H01L2924/1434 (20130101); H01L2924/15311 (20130101); H01L2924/182 (20130101)

Background/Summary

BACKGROUND

Field of the Disclosure

[0001] This disclosure relates generally to die interconnect architectures and packaging.

Description of Related Art

[0002] Die packages often include dies, such as chiplets, embedded within a substrate. Die packages may be used across a multitude of applications, such as telecommunication, automotive, cloud-based, gaming, enterprise, and networking applications, among various other applications. The dies within a die package often require connections to die package pins, often referred to as input/output (I/O) pins, to receive or transmit signals. The signals may be received from or transmitted to other die packages. As the number of dies within a die package increases, the number of connections required to I/O pins increases. Similarly, as the number of features supported by dies increase, the number of connections required by those dies to die package pins tends to increase. To accommodate the additional I/O pins required, the size of these die packages are increased. Increases to die package sizes can result in higher costs, such as higher manufacturing costs, and further result in the die packages requiring larger real estate on printed-circuit-boards (PCBs). As such, there are opportunities to address these and other deficiencies associated with die architectures and packaging.

SUMMARY

[0003] According to one aspect, a die package includes a plurality of pins, a first die, and a second die. The first die is electrically coupled (e.g., connected) to a first pin of a plurality of pins over a first interconnect (e.g., a first connecting structure), and to a second pin of the plurality of pins over a second interconnect. The first die is also electrically coupled to the second die over a third interconnect. The first die is configured to receive a first signal over the first interconnect. Based on the first signal, the first die is configured to electrically couple the second interconnect to the third interconnect. Further, based on the electrical coupling, the first die is configured to receive a second signal over the second interconnect, and transmit the second signal to the second die over the third interconnect.

[0004] According to another aspect, a die includes a plurality of gates electrically coupled to at least a first pin over at least a first interconnect, and to at least a second pin over at least a second interconnect. The plurality of gates are configured to receive a first signal over the first interconnect. Based on the first signal, the plurality are gates are configured to electrically couple the second interconnect to one of a third interconnect and a fourth interconnect.

[0005] According to yet another aspect, a die package includes a plurality of pins, a first interconnect electrically coupled to a first pin of the plurality of pins, a second interconnect electrically coupled to a second pin of the plurality of pins, and a third interconnect electrically coupled to at least a third pin of the plurality of pins. The die package also at least one processor electrically coupled to the first interconnect and the second interconnect. The at least one processor is configured to receive a first signal from the first interconnect and a second signal from the second interconnect. The least one processor is also configured determine a mode value based on the first signal and the second signal. Further, based on the mode value, the at least one processor is configured to transmit a third signal to a control device, the third signal causing the control device to electrically couple the second interconnect to a third interconnect.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0006] FIG. **1** is a block diagram of a die package, according to some implementations;
[0007] FIG. **2** is a block diagram of a die package, according to some implementations;
[0008] FIG. **3** is a block diagram of a system-on-a-chip, according to some implementations;
[0009] FIG. **4**A is a block diagram of a system-on-a-chip, according to some implementations;
[0010] FIG. **4**B illustrates a controller of the system-on-a-chip of FIG. **4**A, according to some implementations;
[0011] FIG. **5** is a block diagram a die package, according to some implementations;
[0012] FIG. **6** is a flowchart of an exemplary process for electrically connecting dies to pins, according to some implementations; and
[0013] FIG. **7** is a flowchart of another exemplary process for electrically connecting dies to pins, according to some implementations.
DETAILED DESCRIPTION
[0014] While the features, methods, devices, and systems described herein may be embodied in various forms, some exemplary and non-limiting embodiments are shown in the drawings, and are described below. Some of the components described in this disclosure are optional, and some implementations may include additional, different, or fewer components from those expressly described in this disclosure.
[0015] The embodiments described herein are directed to improving pin utilization within die architectures. For instance, die packages include pins that can allow for signaling to and from dies within the die package. The pins can connect the die package to, for example, a printed circuit board (PCB), allowing for input/output (I/O) signals between the dies of the die package and other devices, such as other die packages. Conventionally, a die package may allocate a pin to each I/O signal required by each die. This can cause an increase in the cost and size of die packages. To address these and other issues, the embodiments herein may allow the sharing of pins among multiple dies, thereby reducing the number of pins required by a die package, among other advantages.
[0016] For instance, in some examples, a plurality of pins of a die package are electrically connected to control logic. The control logic is connected to a plurality of dies with corresponding connecting structures, such as bond wires. A portion of the plurality of pins, such as one, two, or four pins, are control pins and define a control mode. The control logic determines the control mode based on signals received from the control pins. Based on the control mode, the control logic electrically connects one or more of the plurality of pins (e.g., any of the non-control pins) to connecting structures of a corresponding die.
[0017] In some instances, the connecting structures between the control logic and the plurality of dies may be shared. For example, the connecting structures may include an enable line for each of the dies as well as shared lines that are shared between two or more of the dies. Based on the control mode, the control logic may select one of the plurality of dies using the enable lines, and may connect one or more of the pins to the shared lines. The selected enable line indicates to the selected die that the shared lines can be used by the selected die for signaling (e.g., for receiving or transmitting signals).
[0018] In some examples, certain signals (e.g., standard signals, critical signals, etc.) for one or more dies bypass the control logic and are directly connected (e.g., bonded out) to corresponding pins on the die package. For example, power, ground, clock, and reset signals of a die may be connected to unshared die package pins. The signals that are directly bonded out to a corresponding package pin can vary based on die architecture design, cost, and, application requirements.
[0019] FIG. **1** is a block diagram of a die package **100** (also referred to as an integrated circuit

package) that includes various dies, such as first die **102**, second die **112**, and third die **122**, within a substrate **101**. For instance, each of the first die **102**, second die **112**, and third die **122** may include an integrated circuit fabricated on a wafer, where the wafer may be encapsulated by an encapsulant. In some examples, one or more of the first die **102**, second die **112**, and third die **122** may be a field-programmable gate array (FPGA), an application-specific integrated circuit (ASIC), a state machine, digital circuitry, or any other suitable circuitry or hardware.

[0020] The die package **100** may also include a plurality of package pins **130**. The package pins **130** may be any suitable connecting structure, such as pins or metal pads, for example. In this example, each of the plurality of package pins **130** may be electrically connected to a corresponding solder ball **131**. For instance, the package pins **130** and corresponding solder balls **131** may form a ball grid array (BGA). The solder balls **131** may facilitate electrical connections to a printed circuit board (PCB) **150**. For example, to place and secure to the PCB **150**, the solder balls **131** may be heated, causing the solder balls **131** to melt. The solder balls **131** are then allowed to cool and solidify, thereby forming soldered connections between the substrate **101** and the PCB **150**.

[0021] In this example, first die **102** includes a plurality of first pads **103** that allow for electrical connections (e.g., signaling) to and from the first die **102**. For example, the first pads **103** may be metal pads through which electrical connections are provided. In some instances, the first pads **103** may include metallic flip-chip bumps through which the electrical connections are provided. Further, one or more of the first pads **103** may be electrically connected via a first interconnect **105** to a corresponding package pin **130**. The first interconnects **105** may be any suitable connecting structure, such as a substrate trace (e.g., substrate route) or bond wire. For example, first pad **103**A may be electrically connected to package pin **130**A through first interconnect **105**A. Similarly, first pad **103**B may be electrically connected to package pin **130**B through first interconnect **105**B. In some examples, two or more first pads **103** are electrically connected to a same package pin **130**. For instance, first pads **103**C and **103**D may be electrically connected to package pin **130**C through first interconnect **105**C.

[0022] Additionally, second die **112** includes a plurality of second pads **113** that may be electrically connected via second interconnects **115** (e.g., substrate traces, bond wires) to a corresponding first pad **103** of the first die **102**. For instance, as illustrated, second pad **113**A of second die **112** may be electrically connected to first pad **103**D of first die **102** via second interconnect **115**A. Similarly, second pad **113**F of the second die **112** may be electrically connected to first pad **103**G of the first die **102** via second interconnect **115**D. In some examples, two or more of the second pads **113** of the second die **112** may be electrically connected to a same first pad **103** of the first die **102**. For example, each of the second pads **113**B, **113**C of the second die **112** may be electrically connected to first pad **103**E of the first die **102** via second interconnect **115**B. Similarly, each of the second pads **113**D, **113**E of the second die **112** may be electrically connected to first pad **103**F of the first die **102** via second interconnect **115**C.

[0023] Further, third die **122** includes a plurality of third pads **123** that may be electrically connected via third interconnects **125** (e.g., substrate traces, bond wires) to a corresponding first pad **103** of the first die **102**. For example, third pad **123**A of third die **122** may be electrically connected to first pad **103**H of first die **102** via third interconnect **125**A. In addition, third pad **123**D of third die **122** may be electrically connected to first pad **103**J of first die **102** via third interconnect **125**C, and third pad **123**E of third die **122** may be electrically connected to first pad **103**K of first die **102** via third interconnect **125**D. In some examples, two or more of the third pads **123** of the third die **122** may be electrically connected to a same first pad **103** of the first die **102**. For example, as illustrated, third pads **123**B, **123**C of the third die **122** may be electrically connected to the first pad **1031** of the first die **102** via third interconnect **125**B.

[0024] The first die **102** includes control logic **104** that is configured to route signals received on at least a portion of the plurality of package pins **130** to the first die **102** itself, the second die **112**, and the third die **122**. For instance, and as described herein, control logic **104** may determine a control mode based on signals received from corresponding ones of the plurality of package pins **130**. The signals may be provided by another device, such as another die package, that is electrically connected to at least package pins **130**A, **130**B (e.g., via tracing on PCB **150**). For instance, the other device may provide the control mode on package pins **130**A, **130**B to provide signaling to, or receive signaling from, die package **100**. Based on the control mode, the control logic **104** may electrically connect (e.g., electrically enable) one or more of the plurality of package pins **130** to additional logic within the first die **102**, to the second die **112**, or to the third die **122**.

[0025] In one example, signals received on two pins, such as package pins **130**A, **130**B, define the control mode. These package pins **130**A, **130**B may be referred to as control pins. The control logic **104** may include logic gates that, based on signals received from package pins **130**A, **130**B (e.g., via first pads **103**A, **103**B), electrically connect at least a portion of the remaining package pins **130**, such as package pins **130**C, **130**D, **130**E, **130**F, to one of the first die **102**, second die **112**, or third die **122**. As an example, when the signals received from the package pins **130**A, **130**B are each "low" (e.g., **0** Volts when active high), control logic **104** may electrically connect package pins **130**C, **130**D, **130**E, **130**F to additional logic within first die **102**, such as function logic **106**, to allow for the execution of one or more functions of the first die **102**.

[0026] When, however, the signal received from the package pin **130**A is "low," and the signal received from the package pin **130**B is "high" (e.g., 5 Volts when active high), control logic **104** may electrically connect package pins **130**C, **130**D, **130**E, **130**F to the second die **112**. For instance, control logic **104** may electrically connect package pin **130**C to second interconnect **115**A, package pin **130**D to second interconnect **115**B, package pin **130**E to second interconnect **115**C, and package pin **130**F to second interconnect **115**D.

[0027] Further, when the signal received from the package pin **130**A is "high," and the signal received from the package pin **130**B is "low," control logic **104** may electrically connect package pins **130**C, **130**D, **130**E, **130**F to the third die **122**. For instance, control logic **104** may electrically connect package pin **130**C to third interconnect **125**A, package pin **130**D to third interconnect **125**B, package pin **130**E to third interconnect **125**C, and package pin **130**F to third interconnect **125**D.

[0028] Although, in this example, control logic **104** is described as electrically connecting four pins, namely package pins **130**C, **130**D, **130**E, **130**F, to one of first die **102**, second die **112**, and third die **122**, in other examples control logic **104** may electrically connect less than, or more than, four pins to any number of dies. Moreover, although two package pins, namely package pins **130**A, **130**B, in this example define the control mode, in other examples additional package pins may be used to define the control mode.

[0029] FIG. **2** is a block diagram of a die package **200** that includes a first die **202**, a second die **212**, and a third die **222** embedded within a substrate **201** (e.g., mold compound). The die package **200** further includes package pins **230** (e.g., solder balls) to facilitate attachment to an assembly, such as a PCB. The first die **202** includes flip-chip bumps **203** that allow for signaling to and from integrated circuits of the first die **202**. In addition, die package **200** includes interconnects **205** (e.g., substrate routing) that connect one or more of the flip-chip bumps **203** to a corresponding package pin **230**. For example, interconnect **205**A connects package pin **230**A to flip-chip bump **203**A, and interconnect **205**B connects package pin **230**B to flip-chip bump **203**B.

[0030] First die **202** includes control logic **204** that is configured to electrically switch access to one or more of the flip-chip bumps **203** based on a control mode defined by signals received from package pins **230**D and **230**E. For example, based on a level (e.g., low, high) of each signal received on package pins **230**D, **230**E, control logic **204** may electrically switch access to package pins **230**A and **230**B to one of the first die **202**, second die **212**, and third die **222**. When, for instance, the control mode indicates a selection of the second die **212** (e.g., signal from package pin **230**D is low, and signal from package pin **230**E is high), the control logic **204** may electrically connect the package pins **230**A, **230**B to corresponding interconnects **260**A (e.g., bond wires, substrate routing), which are electrically connected to second die **212**.

[0031] When, however, the control mode indicates a selection of the third die **222** (e.g., signal from package pin **230**D is high, and signal from package pin **230**E is low), the control logic **204** may electrically connect the package pins **230**A, **230**B to corresponding interconnects **260**B (e.g., bond wires, substrate routing), which are electrically connected to third die **222**. Further, when the control mode indicates a selection of the first die, (e.g., signal from package pin **230**D is low, and signal from package pin **230**E is low), the control logic **204** may electrically connect the package pins **230**A, **230**B to one or more logic gates (e.g., an integrated circuit) of the first die **202**.

[0032] Further, in this example, third die **222** includes connections to package pins **230** that bypass control logic **204**. For instance, bond wire **243** connects third die **222** to interconnect **205**B, which is electrically connected to package pin **230**B. In addition, bond wire **240** connects third die **222** to interconnect **205**C, which is electrically connected to package pin **230**C. In some examples, package pins **230**B, **230**C may provide critical signals, such as power, ground, or other signals. As such, in this example, the signals provided from package pins **230**B, **230**C are provided directly to third die **222**, bypassing the control logic **204** of the first die **202**.

[0033] In addition, and to facilitate die-to-die signaling that bypasses control logic **204**, die package **200** may include interconnects **205** connecting one die to another. For example, interconnect **205**D connects flip-chip bump **203**C directly to bond wire **242**, which is electrically connected to second die **212**. An integrated circuit of first die **202** with access to flip-chip bump **203**C may, for instance, receive a signal from, or transmit a signal to, an integrated circuit of the second die **212** over interconnect **205**D.

[0034] FIG. **3** is a block diagram of a system-on-a-chip (SoC) **300** that includes a first die **302**, a second die **312**, and a third die **322**. In this example, first die **302** is electrically coupled to package pins **330** via connecting structures **329** (e.g., substrate tracing, bond wires). Based on signals received from a predetermined number (e.g., one, two) of the package pins **330**, controller **304** of the first die **302** is configured to electrically couple at least a portion of the remaining package pins **330** to one of second die **312** and third die **322**. For example, the signals received from two of the package pins **330** can define a control mode. The control mode may identify an intended connection for the remaining package pins **330**. For instance, a control mode of 0b00 (e.g., as defined by a level of the two package pins **330**) may indicate a connection to first die **302**. A control mode of 0b01 may indicate a connection to second die **312**, and a control mode of 0b10 may indicate a connection to third die **322**.

[0035] When the control mode indicates a connection to the first die **302**, the controller **304** electrically connects the remaining pins **330** to internal circuitry of the first die **302**. When the control mode indicates a connection to the second die **312**, the controller **304** electrically connects the remaining pins **330** to interconnect bus **303** (e.g., one or more bond wires), which is electrically connected to the second die **312**. Further, when the control mode indicates a connection to the third die **322**, the controller **304** electrically connects the remaining pins **330** to interconnect bus **305**, which is electrically connected to the third die **322**. As such, at least a portion of the plurality of pins are shared (e.g., mode-based sharing) between the first die **302**, second die **312**, and third die **322** for signaling.

[0036] As illustrated, SoC **300** may, in some examples, include an interconnect bus **307** that electrically connects the first die **302** to the second die **312**. The interconnect bus **307** is independent of any switching operations performed by the controller **304**, and allows for die-to-die direct signaling between the first die **302** and the second die **312**. Similarly, SoC **300** may, in some examples, include an interconnect bus **311** that electrically connects the first die **302** to the third die **322**. The interconnect bus **311** is independent of any switching operations performed by the controller **304**, and allows for die-to-die direct signaling between the first die **302** and the third die **322**. SoC **300** may further include, in some examples, an interconnect bus **309** that electrically connects the second die **312** to the third die **322**. For instance, second die **312** may receive a signal from, or transmit a signal to, the third die **322** over the interconnect bus **309**.

[0037] FIG. **4**A is a block diagram of a SoC **400** that includes a first die **402**, a second die **412**, and a third die **422**. Each of the first die **402**, second die **412**, and third die **422** may include an integrated circuit, for example. As illustrated, first die **402** includes a controller **404** that is electrically coupled to function logic **406** over interconnect bus **407**. The controller **404** is also electrically coupled to control pins **401**A, **401**B, and I/O pins **413**A, **413**B (e.g., data pins). Although only two control pins **401**A, **401**B, and two I/O pins **413**A, **413**B, are illustrated, other examples can include any suitable number of control pins (e.g., based on the number of dies of a structure), and any suitable number of I/O pins (e.g., based on the highest number of pins required by any one of the dies of the structure). Further, the controller **404** is electrically coupled to the second die **412** over interconnect bus **405**, and to the third die over interconnect bus **409**.

[0038] Controller **404** is configured to receive a first control signal from control pin **401**A and a second control signal from control pin **401**B. The first control signal and second control signal may be received from another device (e.g., another SoC) or die (e.g., second die **412**, third die **422**), and define a control mode for the I/O pins **413**A, **413**B. For instance, the control mode may be based on a property of the first control signal and the second control signal, such as a signal level (e.g., low, high), frequency, or any other suitable property of each of the first control signal and second control signal. The control modes may include a first control state (e.g., first control signal is low, second control signal is low), a second control state (e.g., first control signal is low, second control signal is high), and a third control state (e.g., first control signal is high, second control signal is high).

[0039] The controller **404** includes logic that, based on the control mode, electrically connects the I/O pins **413**A, **413**B to one of function logic **406**, second die **412**, and third die **422**. For example, if the first control signal and second control signal indicate the first control state, the controller **404** electrically connects the I/O pins **413**A, **413**B to interconnect bus **407**, which is electrically connected to function logic **406**. If, however, the first control signal and second control signal indicate the second control state, the controller **404** electrically connects the I/O pins **413**A, **413**B to interconnect bus **405**, which is electrically connected to the second die **412**. When the first control signal and second control signal indicate the third control state, the controller **404** electrically connects the I/O pins **413**A, **413**B to interconnect bus **409**, which is electrically connected to the third die **422**.

[0040] FIG. **4**B illustrates an example of the controller **404** of FIG. **4**A. As illustrated, the control pins **401**A, **401**B are electrically coupled to a first AND gate **452**, a second AND gate **462**, and a third AND gate **472**. In this example, the first AND gate **452** performs a logical negation (“NOT”) on each of its inputs, namely, the first control signal from control pin **401**A and the second control signal from control pin **401**B, before providing a first AND output signal **453**. The second AND gate **462** performs a logical negation on the first control signal from the control pin **401**A, but not on the second control signal from the control pin **401**B, before providing a second AND output signal **463**. Further, the third AND gate **472** performs a logical negation on the second control signal from the control pin **401**B, but not on the first control signal from the control pin **401**A, before providing a third AND output signal **473**.

[0041] The controller **404** further includes a first data signal buffer **454**, a second data signal buffer **456**, a third data signal buffer **464**, a fourth data signal buffer **466**, a fifth data signal buffer **474**, and a sixth data signal buffer **476**. Each of these buffers may be, for example, tri-state digital buffers. As illustrated, the first data signal buffer **454** receives the first AND output signal **453** as a control input. The control input can “activate” the first data signal buffer **454**, causing the first data signal buffer **454** to electrically connect the I/O pin **413**A to interconnect **455** of interconnect bus **407**. For example, and assuming the first data signal buffer **454** is “active high,” when the first AND output signal **453** is “low” (e.g., when one of the first control signal and second control signal are “high”), the first data signal buffer **454** electrically isolates the I/O pin **413**A from the interconnect **455**. When, however, the first AND output signal **453** is “high” (e.g., when each of the first control signal and second control signal are “low”), the first data signal buffer **454** electrically connects the I/O pin **413**A to the interconnect **455**.

[0042] Similarly, the second data signal buffer **456** receives the first AND output signal **453** as a control input that can “activate” the second data signal buffer **456**, causing the second data signal buffer **456** to electrically connect the I/O pin **413**B to interconnect **457** of interconnect bus **407**. For example, and assuming the second data signal buffer **456** is “active high,” when the first AND output signal **453** is “low,” the second data signal buffer **456** electrically isolates the I/O pin **413**B from the interconnect **457**. When, however, the first AND output signal **453** is “high,” the second data signal buffer **456** electrically connects the I/O pin **413**B to the interconnect **457**.

[0043] Further, the third data signal buffer **464** receives the second AND output signal **463** as a control input that can “activate” the third data signal buffer **464**, causing the third data signal buffer **464** to electrically connect the I/O pin **413**A to interconnect **465** of interconnect bus **405**. For example, and assuming the third data signal buffer **464** is “active high,” when the second AND output signal **463** is “low” (e.g., when the first control signal is “high” or the second control signal is “low”), the third data signal buffer **464** electrically isolates the I/O pin **413**A from the interconnect **465**. When, however, the second AND output signal **463** is “high” (e.g., when the first control signal is “low” and the second control signal is “high”), the third data signal buffer **464** electrically connects the I/O pin **413**A to the interconnect **465**.

[0044] Similarly, the fourth data signal buffer **466** receives the second AND output signal **463** as a control input that can “activate” the fourth data signal buffer **466**, causing the fourth data signal buffer **466** to electrically connect the I/O pin **413**B to interconnect **467** of interconnect bus **405**. For example, and assuming the fourth data signal buffer **466** is “active high,” when the second AND output signal **463** is “low,” the fourth data signal buffer **466** electrically isolates the I/O pin **413**B from the interconnect **467**. When, however, the second AND output signal **463** is “high,” the fourth data signal buffer **466** electrically connects the I/O pin **413**B to the interconnect **467**.

[0045] Additionally, the fifth data signal buffer **474** receives the third AND output signal **473** as a control input that can “activate” fifth data signal buffer **474**, causing the fifth data signal buffer **474** to electrically connect the I/O pin **413**A to interconnect **475** of interconnect bus **409**. For example, and assuming the fifth data signal buffer **474** is “active high,” when the third AND output signal **473** is “low” (e.g., when the first control signal is “low” or the second control signal is “high”), the fifth data signal buffer **474** electrically isolates the I/O pin **413**A from the interconnect **475**. When, however, the third AND output signal **473** is “high” (e.g., when the first control signal is “high” and the second control signal is “low”), the fifth data signal buffer **474** electrically connects the I/O pin **413**A to the interconnect **475**.

[0046] Similarly, the sixth data signal buffer **476** receives the third AND output signal **473** as a control input that can “activate” the sixth data signal buffer **476**, causing the sixth data signal buffer **476** to electrically connect the I/O pin **413**B to interconnect **477** of interconnect bus **409**. For example, and assuming the sixth data signal buffer **476** is “active high,” when the third AND output signal **473** is “low,” the sixth data signal buffer **476** electrically isolates the I/O pin **413**B from the interconnect **477**. When, however, the sixth data signal buffer **476** is “high,” the sixth data signal buffer **476** electrically connects the I/O pin **413**B to the interconnect **477**.

[0047] FIG. **5** is a block diagram of a die package **500** that includes a controller SoC **502**, a first slave SoC **520**, a second slave SoC **522**, and a third slave SoC **524**. As illustrated, controller SoC **502** includes at least one processor **508** communicatively coupled to a memory device **504**, request logic **506**, and control logic **510**. The one or more processors **508** may include, for example, a microcontroller, a central processing unit (CPU), a graphical processing unit (GPU), a digital signal processor (DSP), or any other suitable processing device. The memory device **504** may include, for example, a volatile or non-volatile memory or storage device, such as, for example, random access memory (RAM), static RAM (SRAM), dynamic RAM (DRAM), read-only memory (ROM), erasable programmable ROM (EPROM), electrically erasable programmable ROM (EEPROM), or flash memory, among others. Although illustrated as separate components for ease of description reasons, in some instances, all or portions of the functions performed by request logic **506** and control logic **510** may be performed by a processor, such as processor **508**. In some instances, all or portions of the functions performed by processor **508**, request logic **506**, and control logic **510** are implemented in, and performed by, one or more FPGAs, one or more ASICs, one or more state machines, digital circuitry, any other suitable circuitry, or any suitable hardware.

[0048] Request logic **506** is electrically coupled to each of the first slave SoC **520**, second slave SoC **522**, and third slave SoC **524** via signal traces **521**, **523**, **525**, respectively. For example, each of the signal traces **521**, **523**, **525** may be electrically connected to corresponding control pins **531**, **533**, **535**, which are electrically connected via corresponding interconnects (e.g., substrate routes) to request logic **506**.

[0049] Control logic **510** is electrically coupled to I/O pins **511**, and is configured to electrically connect the I/O pins **511** to each of the first slave SoC **520**, second slave SoC **522**, and third slave SoC **524** via interconnects **581**, **583**, **585**, respectively. For example, processor **508** may generate, and transmit over interconnect bus **563**, one or more control signals to control logic **510**, causing control logic **510** to electrically connect I/O pins **511** to one of the interconnects **581**, **583**, **585**. In some instances, the interconnect bus **563** may include one or more control pins (e.g., control pins **401**A, **401**B) to establish a control mode of the control logic **510**. A first control mode (e.g., 0b00) may cause the control logic **510** to electrically connect I/O pins **511** to interconnect **581**. A second control mode (e.g., 0b01) may cause the control logic **510** to electrically connect I/O pins **511** to interconnect **583**. Further, a third control mode (e.g., 0b10) may cause the control logic **510** to electrically connect I/O pins **511** to interconnect **585**.

[0050] In some examples, processor **508** may determine the control mode based on a mode value **507** stored in the memory device **504**. For example, processor **508** may occasionally (e.g., periodically) read the mode value **507** from the memory device **504**, and generate and transmit the one or more control signals to the control logic based on the mode value **507**.

[0051] In some instances, the mode value **507** may be written to the memory device **504** by the request logic **506**. For example, request logic **506** may receive a request signal from one of the first slave SoC **520**, second slave SoC **522**, and third slave SoC **524** via signal traces **521**, **523**, **525**, respectively, requesting access to I/O pins **511** of die package **500**. Based on the request signal received, request logic **506** may write a corresponding mode value **507** to the memory device **504**. For instance, request logic **506** may write a first mode value **507** (e.g., 0b00) to the memory device **504** when receiving a request signal from the first slave SoC **520**. Similarly, request logic **506** may write a second mode value **507** (e.g., 0b01) to the memory device **504** when receiving a request signal from the second slave SoC **522**, and a third mode value **507** (e.g., 0b10) to the memory device **504** when receiving a request signal from the third slave SoC **524**.

[0052] In some examples, processor **508** may receive an update signal **591** (e.g., interrupt) from the request logic **506** in response to receiving a request signal from any of the first slave SoC **520**, second slave SoC **522**, and third slave SoC **524**. For example, upon updating the mode value **507** within the memory device **504**, the request logic **506** may transmit the update signal **591** to the processor **508**. Based on receiving the update signal **591**, the processor **508** may read the mode value **507** from the memory device, and generate and transmit one or more corresponding control signals to the control logic **510** to cause the control logic **510** to electrically connect the I/O pins **511** to one of the first slave SoC **520**, second slave SoC **522**, and third slave SoC **524**.

[0053] FIG. **6** is a flowchart of an exemplary process **600** for electrically connecting dies to pins, in accordance with some embodiments. Exemplary process **600** may be carried out by a device, such as die package **100**.

[0054] Beginning at block **602**, a first signal is received over a first interconnect electrically coupled to a first pin. For instance, and as described herein, first die **102** may receive a signal from package pin **130**A (e.g., a control pin) over first interconnect **105**A. At block **604**, a second interconnect is electrically connected to either a third interconnect, or a fourth interconnect, based on the first signal. For example, based on a level of a signal received from package pin **130**A, first die **102** can electrically connect one or more of package pins **130**C, **130**D, **130**E, **130**F to one of the first die **102** and the second die **112**. For instance, assuming the received first signal is “high,” control logic **104** may electrically connect first interconnect **1050**, which is electrically coupled to package pin **130**C, to second interconnect **115**A of second die **112**.

[0055] Proceeding to block **606**, a second signal is received from the second interconnect. Based on the connection established at block **604**, the second signal is transmitted on either the third interconnect or the fourth interconnect. For instance, assuming the control logic **104** electrically connected package pin **130**C to second interconnect **115**A of second die **112**, a signal received from package pin **130**C via interconnect **105**C is transmitted on second interconnect **115**A to second die **112**.

[0056] FIG. **7** is a flowchart of an exemplary process **700** for electrically connecting dies to pins, in accordance with some embodiments. Exemplary process **700** may be carried out by a device, such as controller SoC **502**.

[0057] Beginning at block **702**, a first interrupt signal is received. For example, as described herein, request logic **506** may receive a request signal from first slave SoC **520** to request access to I/O pins **511** of controller SoC **502**. In response to the request signal, request logic **506** may update a mode value **507** in the memory device **504**, and may generate an update signal **591** that is received by processor **508**. At block **704**, a mode value is read from a memory device in response to the interrupt signal. For instance, processor **508** may read the updated mode value **507** from the memory device **504**.

[0058] Proceeding to block **706**, a first signal is transmitted to control logic based on the mode value. The first signal causes the control logic to electrically connect a first interconnect to one of a plurality of other interconnects. For instance, as described herein, based on the mode value **507**, processor **508** may transmit one or more control signals to control logic **510**. The one or more control signals cause the control logic **510** to electrically connect I/O pins **511** to one of the interconnects **581**, **583**, **585**.

[0059] Implementation examples are further described in the following numbered clauses:

[0060] 1. A die package comprising:

[0061] a plurality of pins;

[0062] a first die electrically coupled to a first pin of a plurality of pins over a first interconnect, and to a second pin of the plurality of pins over a second interconnect; and

[0063] a second die electrically coupled to the first die over a third interconnect, wherein the first die is configured to: [0064] receive a first signal over the first interconnect; [0065] based on the first signal, electrically couple the second interconnect to the third interconnect; [0066] receive a second signal over the second interconnect and, based on the electrical coupling, transmit the second signal to the second die over the third interconnect.

[0067] 2. The die package of clause 1, comprising a third die electrically coupled to the first die over a fourth interconnect, wherein the first die is configured to:

[0068] receive a third signal over the first interconnect; and

[0069] based on the third signal, electrically couple the second interconnect to the fourth interconnect.

[0070] 3. The die package of clause 2, wherein the first die is configured to receive a fourth signal over the second interconnect and, based on the electrical coupling, transmit the second signal to the third die over the fourth interconnect.

[0071] 4. The die package of any of clauses 1-3, wherein the first die comprises a first gate, and wherein the first signal causes the first gate to activate, the activation causing the second

interconnect to electrically couple to the third interconnect.
[0072] 5. The die package of clause 4, wherein the first die comprises a second gate, and wherein the first signal causes the second gate to deactivate, the deactivation causing the second interconnect to electrically disconnect from a fourth interconnect.
[0073] 6. The die package of any of clauses 1-5, wherein the first die is configured to:
[0074] receive a third signal over a fourth interconnect; and
[0075] based on the third signal, electrically couple the second interconnect to the third interconnect.
[0076] 7. The die package of clause 6, wherein the first die is configured to electrically couple the second interconnect to the third interconnect based on a first level of the first signal and a second level of the third signal.
[0077] 8. The die package of clause 7, wherein the first level is the same as the second level.
[0078] 9. The die package of any of clauses 1-8, wherein the second die is electrically coupled to a third pin of the plurality of pins.
[0079] 10. A die comprising:
[0080] a plurality of gates electrically coupled to at least a first pin over at least a first interconnect, and to at least a second pin over at least a second interconnect, wherein the plurality of gates are configured to: [0081] receive a first signal over the first interconnect; [0082] based on the first signal, electrically couple the second interconnect to one of a third interconnect and a fourth interconnect.
[0083] 11. The die of clause 10, wherein the plurality of gates are configured to:
[0084] electrically couple the second interconnect to the third interconnect based on the first signal;
[0085] receive a second signal over the first interconnect; and
[0086] based on the second signal, electrically couple the second interconnect to the fourth interconnect.
[0087] 12. The die of any of clauses 10-11, wherein the plurality of gates comprises a first gate, and wherein the first signal causes the first gate to activate, the activation causing the second interconnect to electrically couple to the one of the third interconnect and the fourth interconnect.
[0088] 13. The die of clause 12, wherein the plurality of gates a comprises a second gate, and wherein the first signal causes the second gate to deactivate, the deactivation causing the second interconnect to electrically disconnect from another one of the third interconnect and the fourth interconnect.
[0089] 14. The die of any of clauses 10-13, wherein the plurality of gates comprise a first gate and a second gate, and wherein:
[0090] the first gate is configured to: [0091] receive the first signal over the first interconnect; [0092] receive a second signal over a fifth interconnect; and [0093] transmit a third signal to the second gate based on the first signal and the second signal; and
[0094] the second gate is configured to: [0095] receive the third signal as an input; and [0096] in response to the third signal, electrically couple the second interconnect to the one of the third interconnect and the fourth interconnect.
[0097] 15. The die of any of clauses 10-14, wherein the first gate is configured to transmit the third signal to the second gate based on a first level of the first signal and a second level of the second signal.
[0098] 16. A die package comprising:
[0099] a plurality of pins;
[0100] a first interconnect electrically coupled to a first pin of the plurality of pins;
[0101] a second interconnect electrically coupled to a second pin of the plurality of pins;
[0102] a third interconnect electrically coupled to at least a third pin of the plurality of pins; and
[0103] at least one processor electrically coupled to the first interconnect and the second interconnect, wherein the at least one processor is configured to: [0104] receive a first signal from

the first interconnect and a second signal from the second interconnect; [0105] determine a mode value based on the first signal and the second signal; and [0106] based on the mode value, transmit a third signal to a control device, the third signal causing the control device to electrically couple the second interconnect to a third interconnect.

[0107] 17. The die package of clause 16, wherein the at least one processor is configured to:

[0108] receive a fourth signal from the first interconnect and a fifth signal from the second interconnect;

[0109] determine a second mode value based on the fourth signal and the fifth signal; and

[0110] based on the mode value, transmit a sixth signal to the control device, the sixth signal causing the control device to electrically couple the second interconnect to a fourth interconnect.

[0111] 18. The die package of any of clauses 16-17, wherein the control device comprises a first gate, and wherein the third signal causes the first gate to activate, the activation causing the second interconnect to electrically couple to the third interconnect.

[0112] 19. The die package of clause 18, wherein the control device comprises a second gate, and wherein the third signal causes the second gate to deactivate, the deactivation causing the second interconnect to electrically disconnect from a fourth interconnect.

[0113] 20. The die package of any of clauses 18-19, wherein the mode value is a first mode value, and wherein the at least one processor is configured to:

[0114] store the first mode value in a memory device;

[0115] receive a fourth signal from the first interconnect and a fifth signal from the second interconnect;

[0116] determine a second mode value based on the fourth signal and the fifth signal;

[0117] read the first mode value from the memory device and compare the first mode value with the second mode value;

[0118] based on the comparison, determine the second mode value is different from the first mode value; and

[0119] in response to determining that the second mode value is different from the first mode value, transmit a sixth signal to the control device, the sixth signal causing the control device to electrically decouple the second interconnect from the third interconnect, and electrically couple the second interconnect to a fourth interconnect.

[0120] Although the methods described above are with reference to the illustrated flowcharts, many other ways of performing the acts associated with the methods may be used. For example, the order of some operations may be changed, and some embodiments may omit one or more of the operations described and/or include additional operations.

[0121] In addition, the methods and system described herein may be at least partially embodied in the form of computer-implemented processes and apparatus for practicing those processes. The disclosed methods may also be at least partially embodied in the form of tangible, non-transitory machine-readable storage media encoded with computer program code that, when executed, causes a machine to fabricate at least one integrated circuit that performs one or more of the operations described herein. For example, the methods may be embodied in hardware, in executable instructions executed by a processor (e.g., software), or a combination of the two. The media may include, for example, RAMs, ROMs, CD-ROMs, DVD-ROMs, BD-ROMs, hard disk drives, flash memories, or any other non-transitory machine-readable storage medium. When the computer program code is loaded into and executed by a computer, the computer becomes an apparatus for causing a machine to fabricate the integrated circuit. The methods may also be at least partially embodied in the form of a computer into which computer program code is loaded or executed, such that, the computer becomes a special purpose computer for causing a machine to fabricate the integrated circuit. For instance, when implemented on a general-purpose processor, computer program code segments can configure the processor to create specific logic circuits. The methods may alternatively be at least partially embodied in application specific integrated circuits or any

other integrated circuits for performing the methods.
[0122] In addition, terms such as “circuit,” “circuitry,” “logic,” and the like can include, alone or in combination, analog circuitry, digital circuitry, hardwired circuitry, programmable circuitry, processing circuitry, hardware logic circuitry, state machine circuitry, and any other suitable type of physical hardware components. Further, the embodiments described herein may be employed within various types of devices such as networking devices, telecommunication devices, smartphone devices, gaming devices, enterprise devices, storage devices (e.g., cloud storage devices), and computing devices (e.g., cloud computing devices), among other types of devices.
[0123] The subject matter has been described in terms of exemplary embodiments. Because they are only examples, the claimed inventions are not limited to these embodiments. Changes and modifications may be made without departing the spirit of the claimed subject matter. It is intended that the claims cover such changes and modifications.

Claims

1. A die package comprising: a plurality of pins; a first die electrically coupled to a first pin of a plurality of pins over a first interconnect, and to a second pin of the plurality of pins over a second interconnect; and a second die electrically coupled to the first die over a third interconnect, wherein the first die is configured to: receive a first signal over the first interconnect; based on the first signal, electrically couple the second interconnect to the third interconnect; receive a second signal over the second interconnect and, based on the electrical coupling, transmit the second signal to the second die over the third interconnect.
2. The die package of claim 1, comprising a third die electrically coupled to the first die over a fourth interconnect, wherein the first die is configured to: receive a third signal over the first interconnect; and based on the third signal, electrically couple the second interconnect to the fourth interconnect.
3. The die package of claim 2, wherein the first die is configured to receive a fourth signal over the second interconnect and, based on the electrical coupling, transmit the second signal to the third die over the fourth interconnect.
4. The die package of claim 1, wherein the first die comprises a first gate, and wherein the first signal causes the first gate to activate, the activation causing the second interconnect to electrically couple to the third interconnect.
5. The die package of claim 4, wherein the first die comprises a second gate, and wherein the first signal causes the second gate to deactivate, the deactivation causing the second interconnect to electrically disconnect from a fourth interconnect.
6. The die package of claim 1, wherein the first die is configured to: receive a third signal over a fourth interconnect; and based on the third signal, electrically couple the second interconnect to the third interconnect.
7. The die package of claim 6, wherein the first die is configured to electrically couple the second interconnect to the third interconnect based on a first level of the first signal and a second level of the third signal.
8. The die package of claim 7, wherein the first level is the same as the second level.
9. The die package of claim 1, wherein the second die is electrically coupled to a third pin of the plurality of pins.
10. A die comprising: a plurality of gates electrically coupled to at least a first pin over at least a first interconnect, and to at least a second pin over at least a second interconnect, wherein the plurality of gates are configured to: receive a first signal over the first interconnect; based on the first signal, electrically couple the second interconnect to one of a third interconnect and a fourth interconnect.
11. The die of claim 10, wherein the plurality of gates are configured to: electrically couple the

second interconnect to the third interconnect based on the first signal; receive a second signal over the first interconnect; and based on the second signal, electrically couple the second interconnect to the fourth interconnect.

12. The die of claim 10, wherein the plurality of gates comprises a first gate, and wherein the first signal causes the first gate to activate, the activation causing the second interconnect to electrically couple to the one of the third interconnect and the fourth interconnect.

13. The die of claim 12, wherein the plurality of gates a comprises a second gate, and wherein the first signal causes the second gate to deactivate, the deactivation causing the second interconnect to electrically disconnect from another one of the third interconnect and the fourth interconnect.

14. The die of claim 10, wherein the plurality of gates comprise a first gate and a second gate, and wherein: the first gate is configured to: receive the first signal over the first interconnect; receive a second signal over a fifth interconnect; and transmit a third signal to the second gate based on the first signal and the second signal; and the second gate is configured to: receive the third signal as an input; and in response to the third signal, electrically couple the second interconnect to the one of the third interconnect and the fourth interconnect.

15. The die of claim 10, wherein the first gate is configured to transmit the third signal to the second gate based on a first level of the first signal and a second level of the second signal.

16. A die package comprising: a plurality of pins; a first interconnect electrically coupled to a first pin of the plurality of pins; a second interconnect electrically coupled to a second pin of the plurality of pins; a third interconnect electrically coupled to at least a third pin of the plurality of pins; and at least one processor electrically coupled to the first interconnect and the second interconnect, wherein the at least one processor is configured to: receive a first signal from the first interconnect and a second signal from the second interconnect; determine a mode value based on the first signal and the second signal; and based on the mode value, transmit a third signal to a control device, the third signal causing the control device to electrically couple the second interconnect to a third interconnect.

17. The die package of claim 16, wherein the at least one processor is configured to: receive a fourth signal from the first interconnect and a fifth signal from the second interconnect; determine a second mode value based on the fourth signal and the fifth signal; and based on the mode value, transmit a sixth signal to the control device, the sixth signal causing the control device to electrically couple the second interconnect to a fourth interconnect.

18. The die package of claim 16, wherein the control device comprises a first gate, and wherein the third signal causes the first gate to activate, the activation causing the second interconnect to electrically couple to the third interconnect.

19. The die package of claim 18, wherein the control device comprises a second gate, and wherein the third signal causes the second gate to deactivate, the deactivation causing the second interconnect to electrically disconnect from a fourth interconnect.

20. The die package of claim 18, wherein the mode value is a first mode value, and wherein the at least one processor is configured to: store the first mode value in a memory device; receive a fourth signal from the first interconnect and a fifth signal from the second interconnect; determine a second mode value based on the fourth signal and the fifth signal; read the first mode value from the memory device and compare the first mode value with the second mode value; based on the comparison, determine the second mode value is different from the first mode value; and in response to determining that the second mode value is different from the first mode value, transmit a sixth signal to the control device, the sixth signal causing the control device to electrically decouple the second interconnect from the third interconnect, and electrically couple the second interconnect to a fourth interconnect.